MD HASHIBUL HOSSAIN

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RESEARCH PROFILE

Early-career researcher working at the intersection of plant pathology, fungi–heavy metal interactions and bioremediation, with analytical chemistry and geospatial skills and a growing machine-learning and computer-vision practice. MSc (Distinction, CGPA 3.80/4.00) with thesis work on fungal metal tolerance, bioaccumulation and GIS-guided isolate screening. Writes reproducible R and Python workflows, has trained YOLO object-detection models, and is currently a Junior Research Scientist developing image-based detection of fungal pathogens and greenhouse pests. Seeking a PhD at the interface of [biological problem] and [computational method].

1. EDUCATION

MSc (Thesis) in Forestry *Session 2022–23; thesis examination Sept 2025*
Shahjalal University of Science & Technology (SUST), Sylhet, Bangladesh

- **CGPA 3.80/4.00 (A, With Distinction)**; 40 credits; 30 students in cohort; provisional certificate dated 27 April 2026
- Thesis: *Tolerance, bioaccumulation and bioremediation of selected heavy metals in water and soil by Aspergillus niger and Trichoderma asperellum, with screening of metallo-tolerant landfill fungal isolates*
- Supervisor: Dr Mohammed Masum Ul Haque; co-supervisor: Prof. Romel Ahmed
- Coursework: Data Analysis and Visualization in R (A+), Advanced Research Methodology, Forest Soil Management, Natural Resource Economics, Forest Resource Management, Community-Based Natural Resource Management; Dissertation (4 credits) and Thesis (12 credits) both A+

BSc (Honours) in Forestry *Session 2018–19; examinations held 2023*
Shahjalal University of Science & Technology (SUST), Sylhet, Bangladesh

- **CGPA 3.43/4.00 (B+)**; 160 credits; 58 students in cohort; provisional certificate dated 30 June 2024
- Project (FES-425): *In-vitro assessment of heavy metal tolerance of some selected fungi species*; supervisor: Dr Mohammed Masum Ul Haque
- Relevant coursework: Remote Sensing and GIS, Introduction to Computer Applications, Mycology/Forest Pathology and Forest Protection, Genetics and Tree Improvement, Research Methodology, Environmental Impact Assessment and Auditing

2. RESEARCH EXPERIENCE

Professional and Research Positions

Junior Research Scientist *1 July 2026 – Present*
CredoSense, Markham, Canada

- Develop machine-learning and computer-vision models to detect pathogenic fungi from microscopic spore images [dataset size, model type, metric].
- Build object-detection models for greenhouse pests from open-source image datasets, covering curation, annotation and evaluation [dataset size, classes, metric].
- Extract and quantify plant pigments (chlorophyll, carotenoids) using wet-lab and spectrophotometric methods [number of samples/species].
- Digest plant tissue and analyse macro- and micronutrients by ICP-OES [number of elements/samples].

Research Associate *Jan – Dec 2025*
Dept. of Forestry & Environmental Science, SUST, Sylhet, Bangladesh · Supervisor: Dr Mohammed Masum Ul Haque

- Designed and led a fungal bioremediation project on heavy metals, including field sampling and isolation of fungi from a waste dump site.
- Isolated fungi from resin-forming agarwood trees to study fungus–plant associations.
- Ran fungal and bacterial culture work (aseptic technique, sub-culturing, isolation streaking, morphological screening) in laminar-flow cabinets.
- Operated ICP-OES independently (calibration, sample preparation) to quantify heavy metals in aqueous and digested samples, and UV-Vis spectrophotometry for colorimetric and enzyme-activity assays.
- Analysed and visualised data in R (descriptive statistics, hypothesis testing); worked on plant-disease identification and biocontrol of fungal pathogens.
- Presented findings at international conferences and mentored junior researchers.

Research Assistant *Jan – Dec 2024*

Dept. of Forestry & Environmental Science, SUST, Sylhet, Bangladesh · Supervisor: Dr Mohammed Masum Ul Haque

- Contributed to a plant-pathology and heavy-metal-tolerance project: experiment design and planning for plant–pathogen interaction and metal-stress studies.
- Supported literature review and ran routine laboratory protocols with controlled sample handling and data recording.

Intern and Field Researcher *16–30 Jan 2024*

USAID Bangladesh Ecosystems/Protibesh Activity, Khadimnagar National Park · Supervisor: Md. Modinul Ahsan, PhD

- Completed a compulsory internship in protected-area biodiversity management.
- Conducted a mixed-methods field study, with a co-author, on how well co-management office bearers and People's Forum members know the Protected Area Management Rules, 2017.
- Interviewed 22 respondents (19 People's Forum members from 10 villages and 3 co-management committee office bearers) using purposive sampling and a structured questionnaire, supported by field observation and official records (data collection 17–23 Jan 2024).
- Analysed and visualised the data in R (v4.3.1) and examined whether knowledge varied with educational qualification.

Degree Research Projects

MSc Thesis Research *2022–2025*

SUST, Sylhet, Bangladesh · Supervisor: Dr Mohammed Masum Ul Haque; co-supervisor: Prof. Romel Ahmed

- Quantified growth inhibition and tolerance index of 2 fungal species against 4 metals (As(V), Cd(II), Cr(VI), Pb(II)) across concentration series using ImageJ colony measurement; fitted dose–response models for EC50 and tested effects by two-way ANOVA.
- Profiled antioxidant enzymes (catalase, SOD, glutathione reductase) under metal stress and summarised response patterns with PCA and NMDS.
- Measured biomass metal accumulation and removal efficiency in water and soil microcosms.
- Built a GIS multi-criteria site-selection workflow (ArcMap 10.8, Google Earth Engine, R) with Sentinel-2/5P, Landsat LST, MAIAC AOD, VIIRS and SRTM layers (10-year window, 500 m common grid, sensitivity analysis).
- Collected 10 field samples (7 soil, 3 water), measured bioavailable metals by ICP-OES, and isolated **27 morphologically distinct fungi** by serial dilution; shortlisted **7 isolates (DMFT1–7)** viable on 500 ppm Cr/As.

BSc Research Project *2018–2023*

SUST, Sylhet, Bangladesh · Supervisor: Dr Mohammed Masum Ul Haque

- Tested 3 fungi (*Aspergillus niger*, *Rhizopus stolonifer*, *Trichoderma asperellum*) against Pb, Hg and As at 100, 200 and 500 ppm on PDA, measuring growth at 24, 48 and 72 h.
- Calculated tolerance indices and analysed them in R 4.3.2 (Shapiro–Wilk, factorial ANOVA, Tukey HSD).
- Ranked *T. asperellum* most tolerant and proposed *R. stolonifer* as a biomonitoring candidate.

Additional Computational and Remote-Sensing Projects

Machine-Learning & Data-Science Projects (Python) *Dec 2024 – Feb 2025*

Python & Basic Data Science training, IICT-SUST

- Fine-tuned a **YOLO11s** object detector (transfer learning, Ultralytics) for 2-class leaf detection on a Roboflow Universe dataset (1,107 training and 46 validation images, 100 epochs), then validated it, ran inference, and exported it to TensorFlow.js.
- Analysed the effect of temperature and light intensity on plant growth (54 observations) with pandas and seaborn: data cleaning, correlation analysis, linear regression and time-series trend analysis.

Land-Cover Change Analysis, Karnaphuli Range, Kaptai *Aug 2023*

Coursework project, SUST

- Analysed land-cover and land-use change over a ~560 ha forest range using satellite image analysis [add data source, software and time span].

3. PUBLICATIONS, CONFERENCES & POSTERS

Manuscripts in review / publication process (none yet published)

- **Hossain, M.H.**, Haque, M.M.U., Ahmed, R., & Rahman, M.A. *Integrated screening of metal tolerance and short-contact dry-biomass biosorption reveals metal-specific performance in filamentous fungi.* (Manuscript in review/publication process.)

- Aziz, F.B.A., Sultana, R., Masum, K.M., & **Hossain, M.H.** *Potential urban green space suitability appraisal using GIS-based multi-criteria analysis: A case study of North-eastern Sylhet City, Bangladesh.* (Manuscript in review/publication process.)
- Rekha, P.R., Ghose, A., **Hossain, M.H.**, & Masum, K.M. *Urban green spaces as carnivore refugia: A camera-trap study in a biodiversity-rich university campus in Bangladesh.* (Manuscript in review/publication process.)

Conference papers and presentations

- **Hossain, M.H.**, Ahmed, R., & Haque, M.M.U. (2025, May). *Heavy metal tolerance by fungi: A step toward mycoremediation strategy for contaminated soils.* International Conference and Bioscience Carnival, Rangamati Science and Technology University, Bangladesh (16–18 May 2025). DOI: 10.13140/RG.2.2.29154.03525.

Poster presentations

- **Hossain, M.H.** (2023, Aug). *Mycorrhizal Role of Carbon Sequestration in Forest Soil.* "Contemporary Trends in Forestry Research" seminar, Dept. of Forestry & Environmental Science, SUST, 6 Aug 2023.
- **Hossain, M.H.** (2023, Mar). *Bioaccumulation of Heavy Metals by Fungi and Its Possible Effects on the Environment.* International Day of Forests, SUST, 21 March 2023.

Other research outputs (non-peer-reviewed)

- **Hossain, M.H.**, & Eva, N.J. (2024). *A Study on the Knowledge of the Office Bearers and Peoples Forum (PF) Members of Khadimnagar National Park about Protected Area Management Rules, 2017.* BSc internship report (FES 433), SUST; supervised by M. Ahsan, PhD (USAID Ecosystems/Protibesh Activity). Available on ResearchGate.
- **Hossain, M.H.** (2023, Aug). *Image Analysis of Karnaphuli Range.* Coursework report. Available on ResearchGate.

Research software and open code (GitHub: Mraz11)

- **PDS-12-Project.** Python and data-science course projects (Jupyter/Colab): YOLO11 object detection on a custom leaf dataset, and regression analysis of plant growth against environmental factors.
- **dose-time-metallotolerance-fungi.** Reproducible R pipeline (MIT licence) built as analysis practice for the BSc project: tolerance index, growth inhibition, growth-rate slopes, Tukey-lettered contrasts, TI heatmaps, LL.4 dose–response fits, and IC50 with bootstrap 95% CI and censoring.

4. TECHNICAL & ANALYTICAL SKILLS

Computational, Machine Learning & Computer Vision

- Machine learning and computer vision: object detection (YOLO11/Ultralytics), transfer learning and fine-tuning, model validation and inference, Roboflow dataset handling, TensorFlow.js export, microscopy and spore-image analysis, image annotation and dataset curation
- Python (80-hour certified training): pandas, NumPy, Matplotlib, seaborn, Jupyter/Google Colab; R (statistics, visualisation, reproducible figure export)
- Reproducible research workflows: Git/GitHub, scripted one-command pipelines, open-licensed code
- Statistical modelling: linear regression, ANOVA/Tukey, four-parameter log-logistic dose–response fitting, IC50 with bootstrap confidence intervals and censoring, PCA, NMDS
- ImageJ image-based quantification; satellite image and land-cover change analysis
- GIS and remote sensing: ArcMap 10.8 (Spatial Analyst), Google Earth Engine, Sentinel-2/5P, Landsat, MAIAC AOD, VIIRS, SRTM; multi-criteria analysis
- Bioinformatics (bootcamp); NGS concepts (training); i-Tree urban tree inventory

Wet Lab & Analytical Chemistry

- ICP-OES (calibration, sample preparation, QC; 8-day certified training); microwave-assisted digestion; plant nutrient (macro/micro) analysis
- UV-Vis spectrophotometry; chlorophyll and carotenoid extraction; antioxidant enzyme assays (CAT, SOD, GR)
- Fungal and bacterial culture (aseptic technique, laminar-flow cabinets, isolation, serial dilution, morphological screening); −80 °C cryopreservation
- Metal tolerance, bioaccumulation and bioremediation assays (water and soil microcosms); preparation of heavy-metal-amended media
- Plant pathology; biocontrol of fungal pathogens; molecular biology and NGS (training); field sampling

Field & Social-Science Methods

- Structured questionnaire design, purposive sampling, stakeholder interviews, mixed-methods analysis; protected-area policy and co-management; R for survey data visualisation

Research interests: mycoremediation, fungi–heavy metal interactions, plant pathology, mycorrhizal ecology, pathogen detection, plant physiology, geospatial and remote-sensing analysis, machine learning and computer vision

5. TRAININGS, CERTIFICATIONS & WORKSHOPS

- **WEkEO for Climate Change**, Copernicus & Mercator Ocean International, online, 10 & 12 June 2025
- **Earth Observation for Climate Adaptation and Resilience in South Asia** (international training workshop), AIRCAS (Chinese Academy of Sciences), Jagannath University, BRAC University C3ER, 12–13 April 2025
- **Understanding Climate Change and Climate Finance** (workshop), PKSF / Green Climate Fund, SUST, 27 Feb 2025
- **Core Techniques in Molecular Biology and Next-Generation Sequencing (NGS)** (5 days), Central Dogma Lab, Invent Technologies, Dhaka, 15–19 Feb 2025
- **Python Programming and Basic Data Science** (80 h), IICT-SUST / EDGE project, Bangladesh Computer Council, Dec 2024 – Feb 2025 (issued 17 May 2025); course projects on GitHub (PDS-12-Project)
- **ICP-OES** (8 days), Invent Technologies at SUST, 9–16 Sept 2024
- **Bioinformatics Bootcamp 2.0**, CHIRAL Bangladesh, 7 Sept 2023
- **Urban Tree Inventory using i-Tree** (virtual), USFS International Programs (Compass Program) [add date]

6. LEADERSHIP & SERVICE

- **G Studio Wing Chief**, Green Explore Society, SUST (9th Executive Committee), Mar 2023 – Mar 2024
- **Assistant Publication Secretary**, Green Explore Society, SUST (8th Executive Committee), Mar 2022 – Mar 2023
- **Mentoring** of junior researchers during the Research Associate role

7. AWARDS, GRANTS & HONORS

- **MSc awarded With Distinction** (CGPA 3.80/4.00)
- **National Eco Olympiad 2025:** selected for the National Round (certificate of achievement, 25 Feb 2025)
- **Bangladesh Physics Olympiad 2018:** regional round, category A
- **8th Bangladesh Chemistry Olympiad 2017:** preliminary round, Chittagong region (19 Jan 2018)
- **7th Bangladesh Chemistry Olympiad 2016:** preliminary round (13 Jan 2017)
- **Bangladesh Botany Olympiad 2017:** participant (26 Feb 2017)
- **15th Bangladesh Mathematical Olympiad:** regional round, higher secondary (6 Jan 2017)
- Grants and fellowships: none documented [add if any]

8. LANGUAGES

- **English:** IELTS Academic, 3 Sept 2026, Overall 7.5 (CEFR C1): Listening 8.0, Reading 8.5, Writing 6.5, Speaking 6.5
- **Bengali:** native

9. REFEREES

- **Dr Mohammed Masum Ul Haque**, Associate Professor, Dept. of Forestry & Environmental Science, SUST (BSc and MSc supervisor; Research Assistant and Associate supervisor)
- **Prof. Romel Ahmed**, Dept. of Forestry & Environmental Science, SUST (MSc co-supervisor)
- [CredoSense supervisor: name, title and contact]